

next year

genassist and amy pavel's line of work

Accessibility

CIS
700
0-

Announcements

Reminder to submit lecture topics for the last day of class

Announcements

Quiz 4 is next Tuesday. It covers everything from content creation up through accessibility (today's lecture).

UbiComp envisioning technology in

support of our long-term goals
via commodity sensing

Final unit

via infrastructure-mediated sensing

accessibility

ICT4D

Last time

HCI's interdisciplinarity makes available many methodological orientations. Which to apply depends on your goal. To wit:

Systems: engineer a thing

Experiments: prove a causal thing

Ethnography: understand a thing

Design: craft a thing

Computational social science: analyze a thing

(Critical) theory: think a thing

Design your evaluation by starting back at your thesis, mapping that thesis onto a claim, then deriving the evaluation from that claim

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Who gets left out of design?

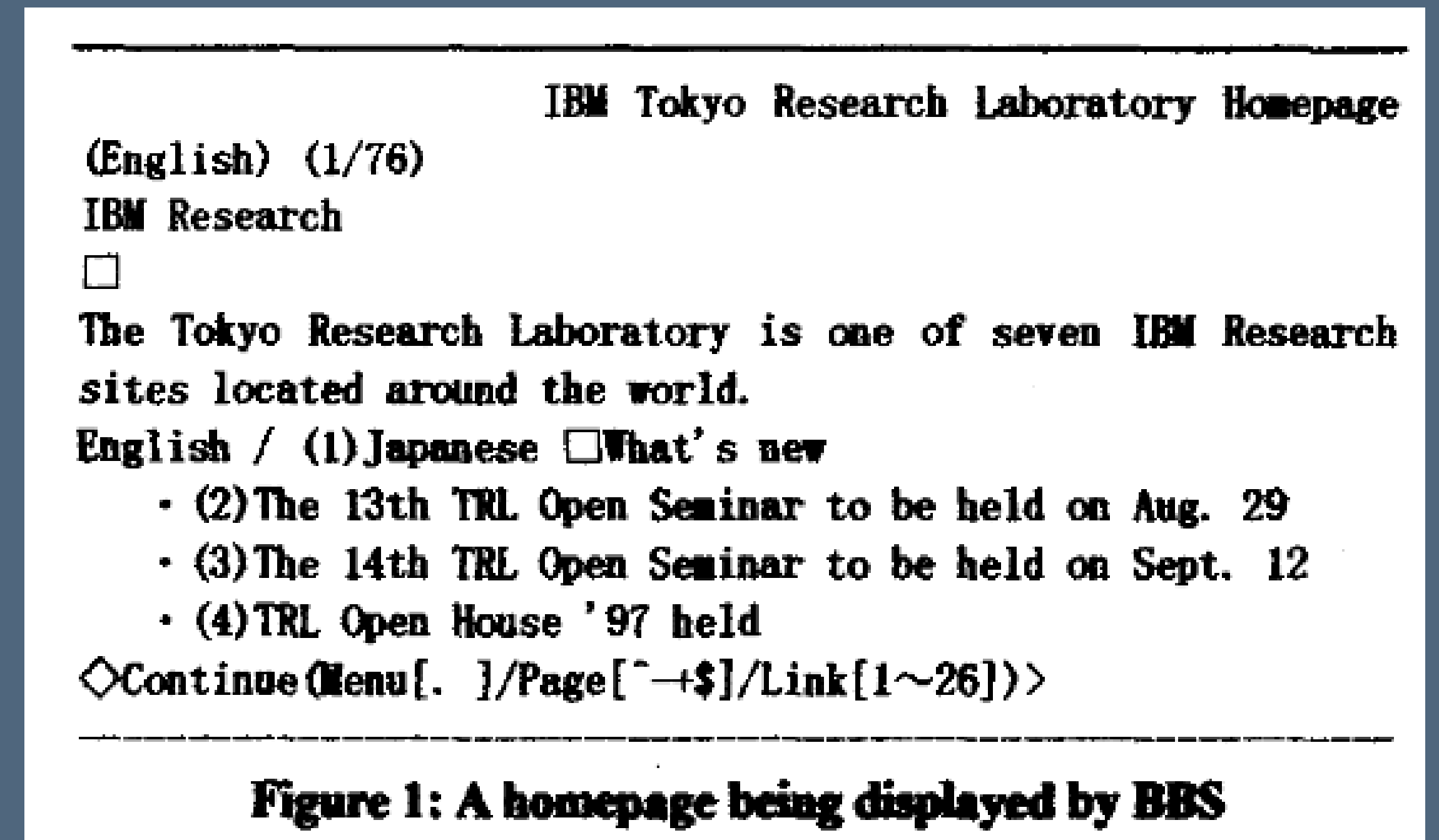
Example. Asakawa and Itoh in 1998 noted challenges of blind web users.

They would use Lynx, where...

they needed to know the keyboard commands

hyperlinks were not easily recognizable

they needed to do a lot of scrolling, as the screen cuts off at 80 columns and 24 lines

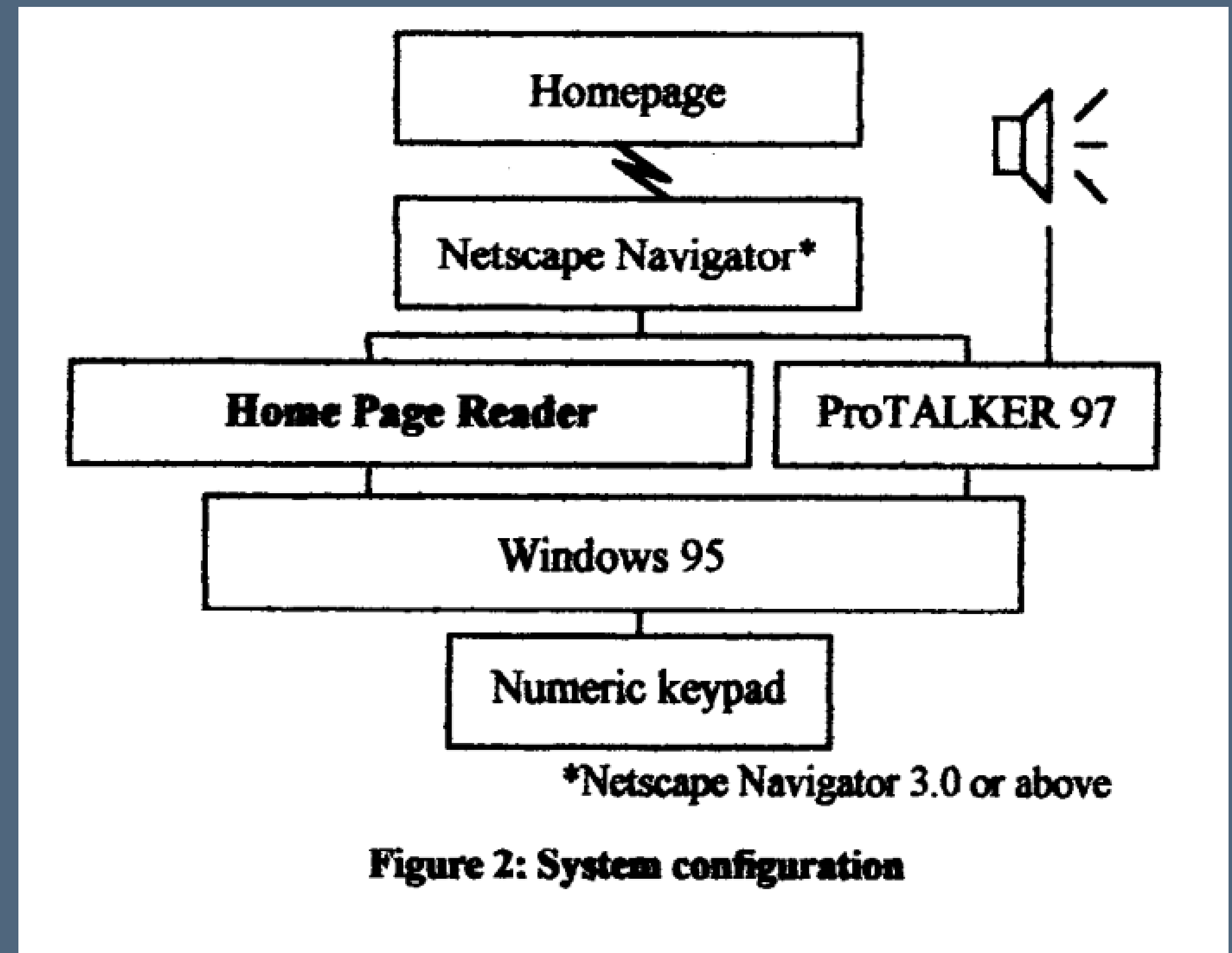


How do we read screens?

[Asakawa and Itoh 1998]

1998: the first interface for reading the web out loud

The challenge: providing audio access to two-dimensional, complex web designs?



Disability is not uncommon

19% of the US population experiences some form of **disability** [US Census Bureau 2012, via Jeff Bigham]

Research targets one or more of many groups: [Jenny Lay-Flurrie]

Visual: colorblind, low vision, blind

Hearing: loss, deaf

Cognitive: dyslexia, seizure, learning disabilities, autism

Speech: speech impediment, unable to speak

Mobility: arthritis, quadripilegia

Neural: bipolar, anxiety, PTSD, depression

Motivating question

Are there general design approaches that could help us make technology that works better for everyone, including those with disabilities who are often forgotten or misrepresented in design?

Today

Ability-based design

Supporting access beyond the screen and individual

Retrofitting accessibility to existing interfaces

~~Ability-based design~~

How disability is talked about

Medical model of disability: the diagnosis is in the individual's own body.

Implication: something is wrong with their body, and quality of life is reduced

Social model of disability: the diagnosis is with behaviors, attitudes, and barriers erected by society, not with the individual.

Physiological differences may produce variation, but it's only disability if society cannot make accommodations

Universal design

General idea: make a single design accessible to everyone.

Example: curb cuts

Credits for these slides: Lloyd May, Stanford PhD, inclusive media technologies, email with questions about accessibility at lloyd.may@monash.edu



Inclusive design practices

Rather than supporting everyone, we focus on individual people who may be underserved, and tailor technology for them.

Pullin and Newell call this “focusing on extra-ordinary users” (2007). They focus on older adults.

Their concerns: universal products are difficult to design. “Accessible” technology is sometimes technically “accessible” but still not very usable.

Instead, **just make good products for different users.**

Bonus: sometimes the designs are useful beyond the group of extra-ordinary users you designed for (cassette recorders, Oxo knives).

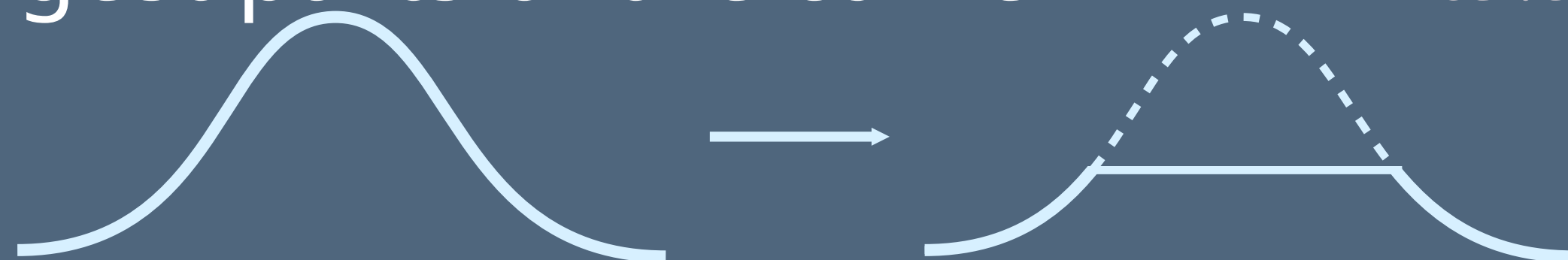
Designing for accessibility

Argument: to design for accessibility, invert the typical design process

Typically, this begins with designing for the common user and then developing accessibility hooks

However, by designing for extreme users, you benefit everybody. Design from the outside in, not the inside out. [Pullin and Newell 2007]

Trevinarus calls this the “lawnmower of justice”: cap effort spent on the largest parts of the curve <https://twitter.com/juttatrevira/status/1105196265216897030>



Recall: Needfinding doesn't solve this

[Bennett and Rosner 2019]

We espouse  as a designer's solution to knowing enough about a stakeholder group to design with or for them

However, typical design strategies  can lend false confidence and incorrect conclusions (e.g., blindfolds)

Ability-based design

[Wobbrock et al. 2011; Wobbrock et al. 2018]

A challenge to the community: **make affordances broadly accessible, but through the specific abilities each user has.**

Not exact
for adap

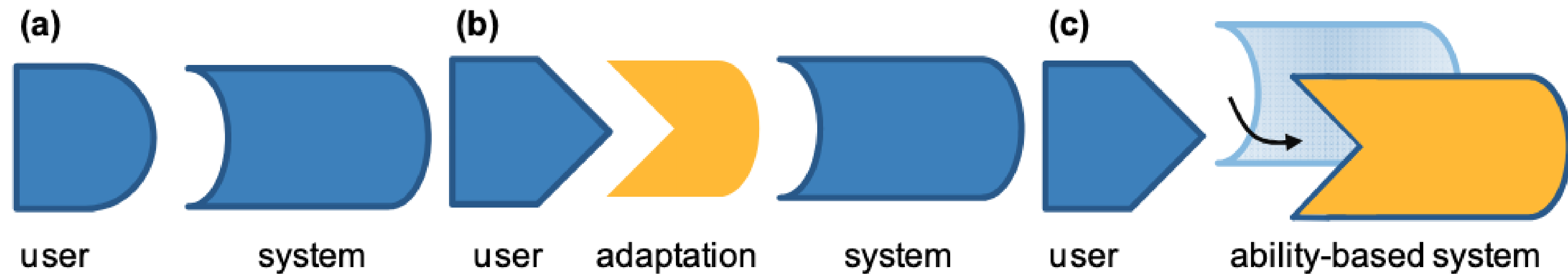


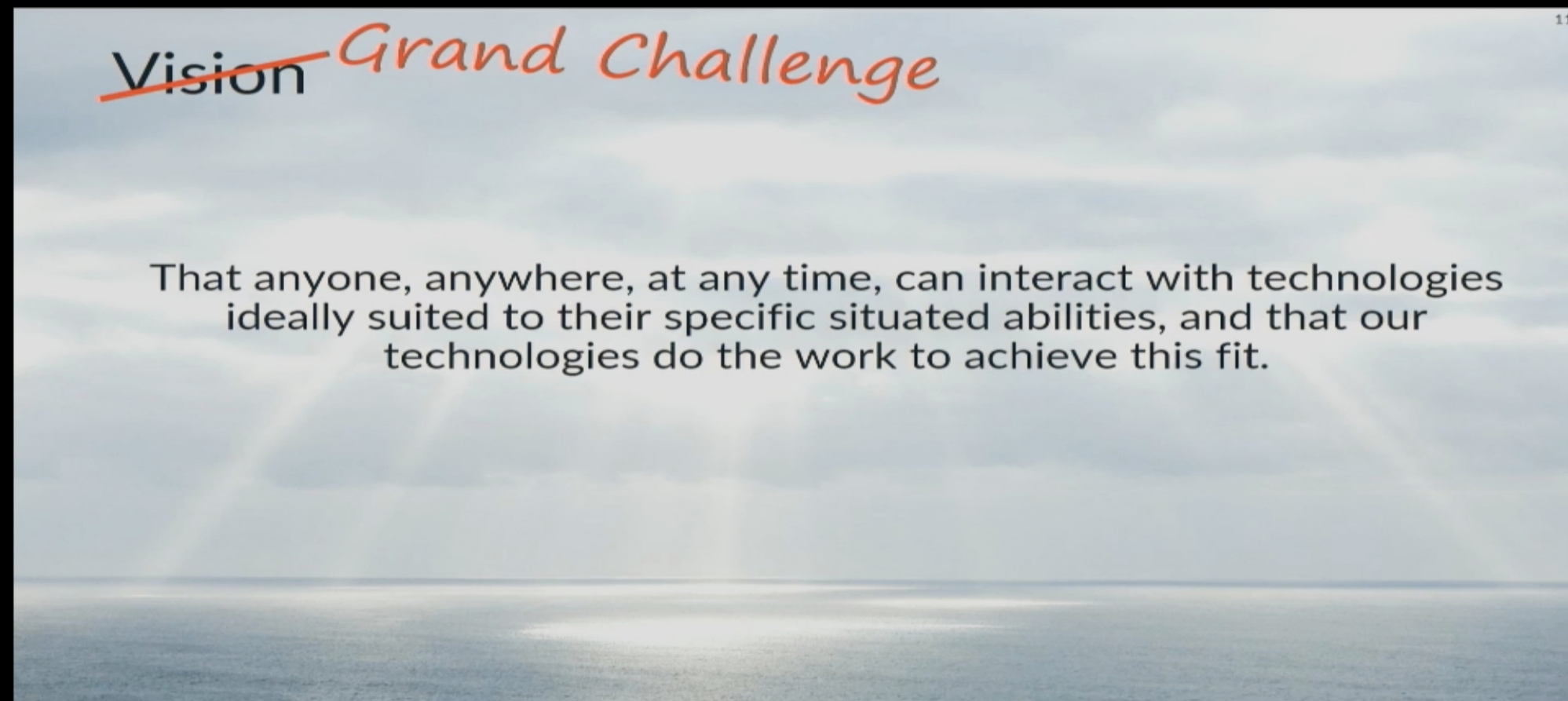
Figure 2. (a) A user whose abilities match those presumed by the system. (b) A user whose abilities do not match those presumed by the system. Because the system is inflexible, the user must be adapted to it. (c) An ability-based system is designed to accommodate the user's abilities. It may adapt or be adapted to them. Our symbols are based on those from prior work (Edwards 1995).

cates
ances.

Ability-based design

[Wobbrock et al. 2011; Wobbrock et al. 2018]

Stop thinking about “dis”-ability, and think about ability-



<https://vimeo.com/218330703>

Where did this come from?

* Andrew's conjecture

You got a flavor for this in examples shared in the ability-based design reading. Around that time, a lot of work was happening in new sensing techniques and new forms of pervasive technology.

My guess? It was becoming more possible to imagine the possibility of accepting more modalities in more contexts. More opportunities and need to capture varied

ability.

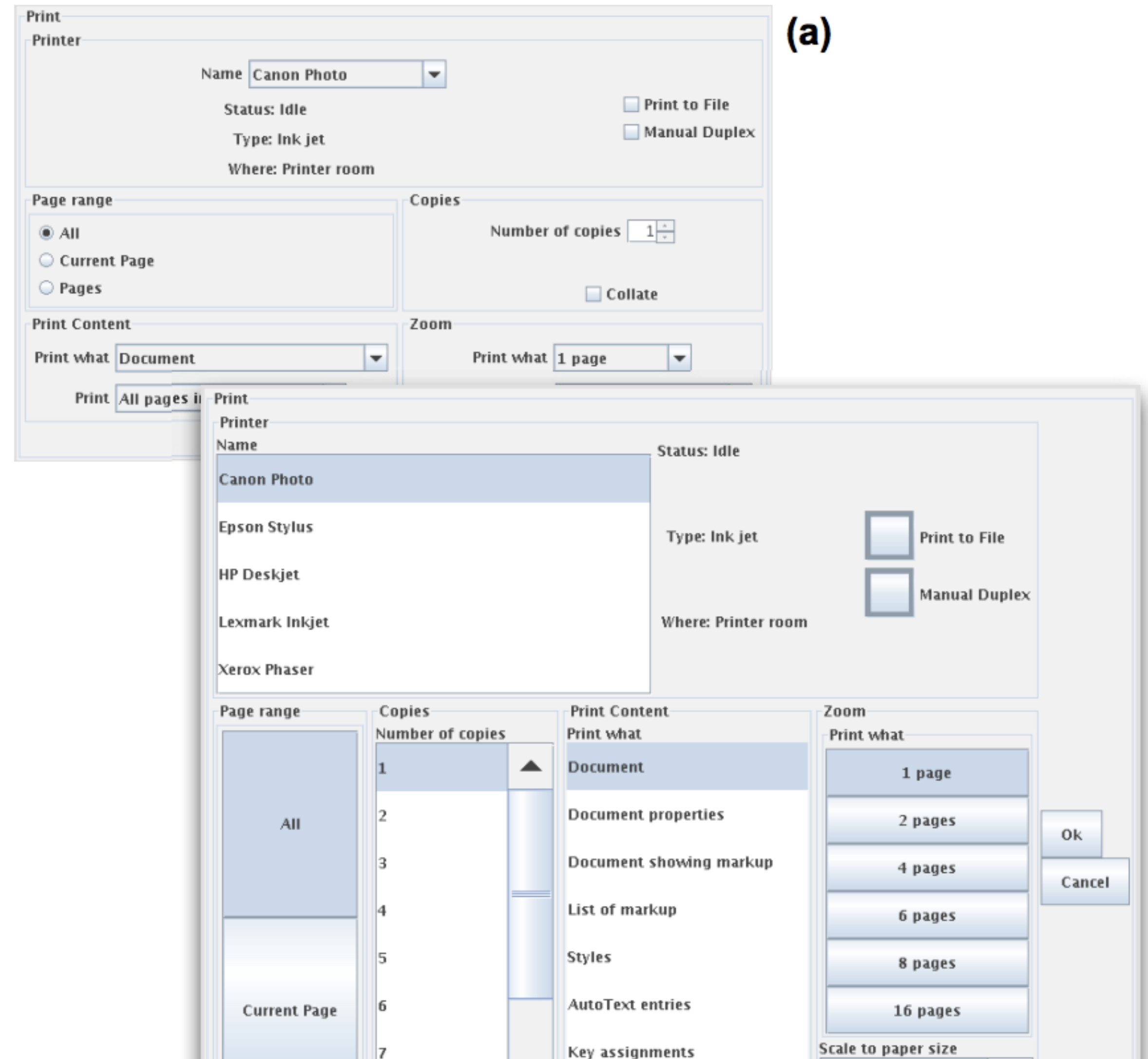
Example: Rearrange the

UI

What if you could rearrange an entire user interface based on a user's abilities?

Step one: model the user's motor abilities

Step two: customize the user interface to match those abilities

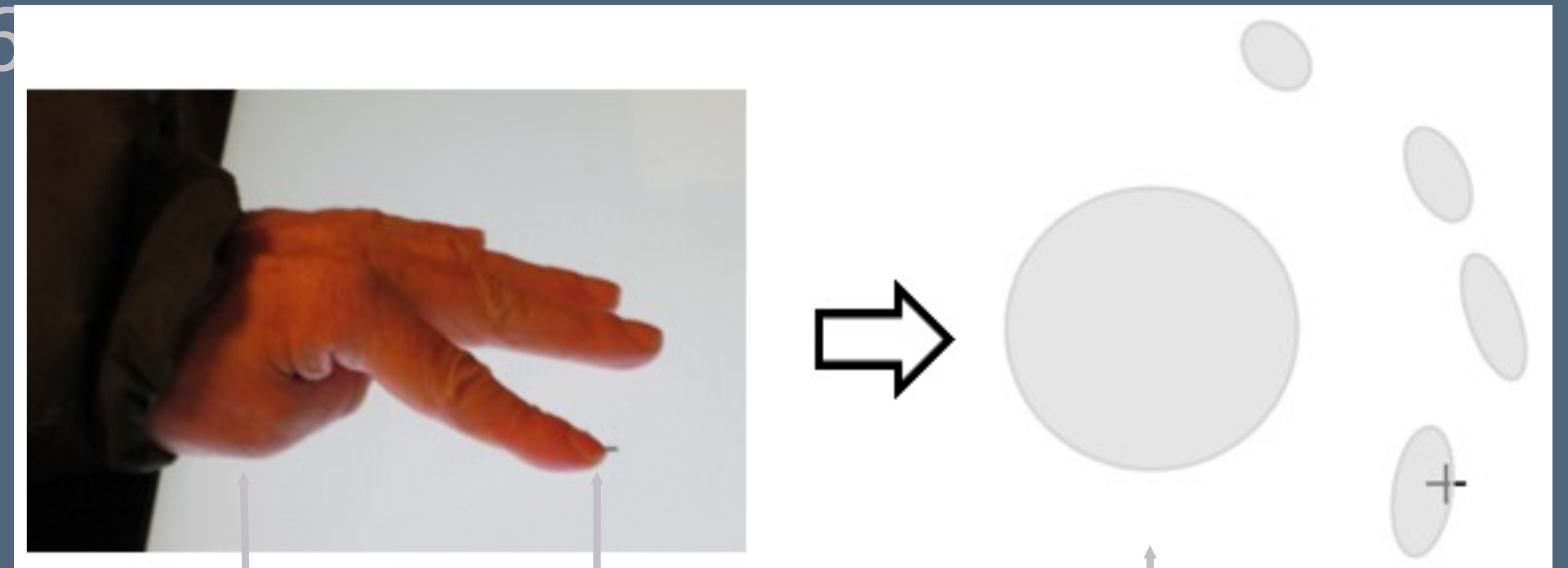


Example: Tolerant decoding

Smart Touch - [Mott et al. '16]

Motor impairments cause touchscreens to register false touches

So: first capture multiple frames of the user's gesture, and find the most stable part with few new up/down points. Then, template match to previous gestures' selected points.



Intended touch point

Unintended touch points, confusing the recognizer

Thinking in abilities

[Wobbrock et al. 2011; Wobbrock et al. 2018]

If you can, think about someone you are close to that has/had a disability. Think about an interaction that they don't have access to.

What are the abilities that they have that relate to that interaction?

How could that interaction be redesigned to support them?

Examples of ability- based design

More focused audio output

[Asakawa and Itoh 1998]

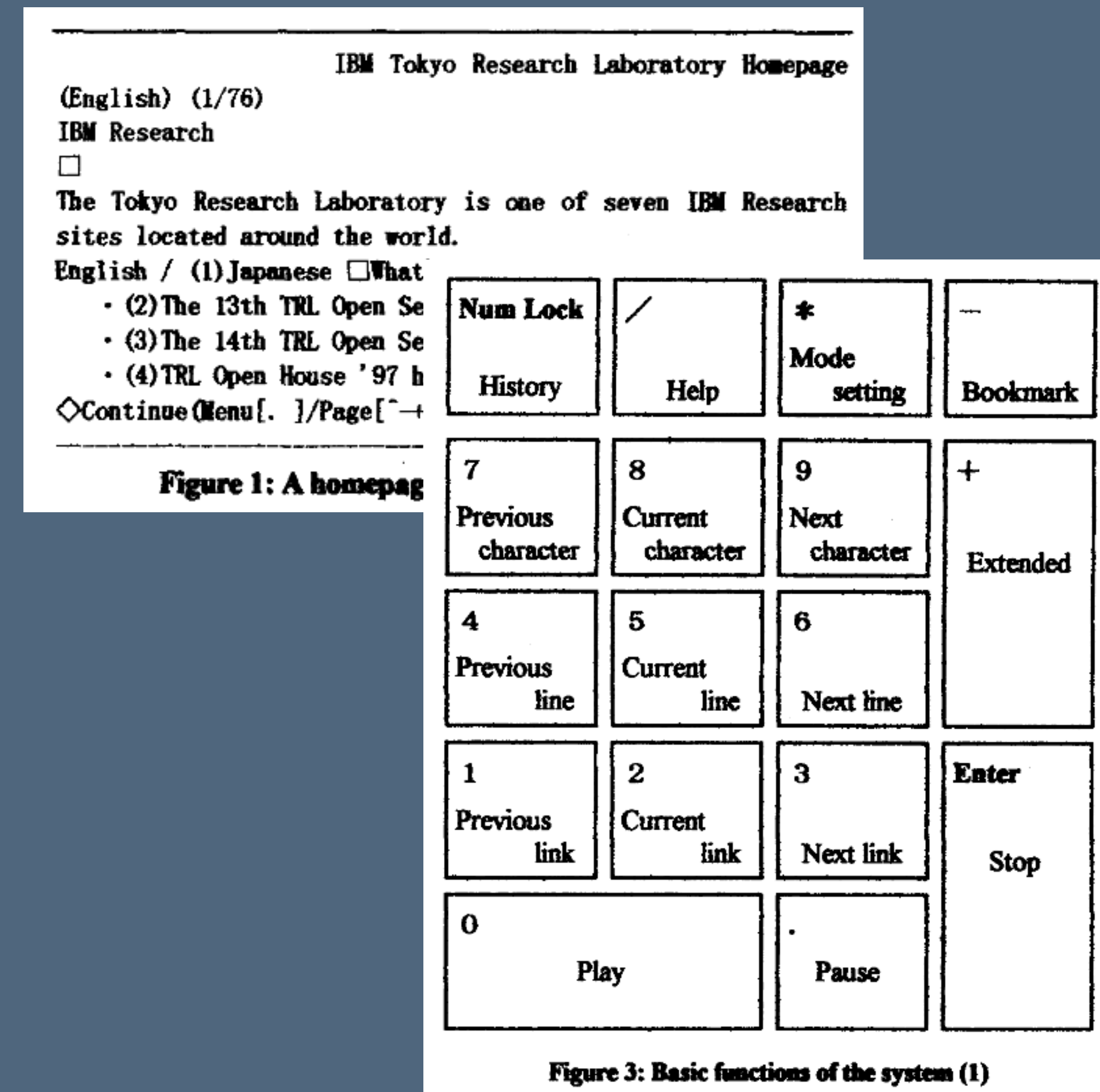
Improved web screen reader interface with a few changes:

provide new key navigation functions

provide a fast-forward key for screen reading

read hyperlinks in a different

voice, read aloud the HTML



Sound awareness via visuals

[Jain et al. 2015]

Gestures and lip movement help interpret speech, but hearing aids and cochlear implants cannot help the listener identify the sound source

Could augmented reality help a person identify a sound's source?

Using expressivity of touch

[Kane, Bigham and Wobbrock 2008]

How might blind users
interact with touchscreen
phones?

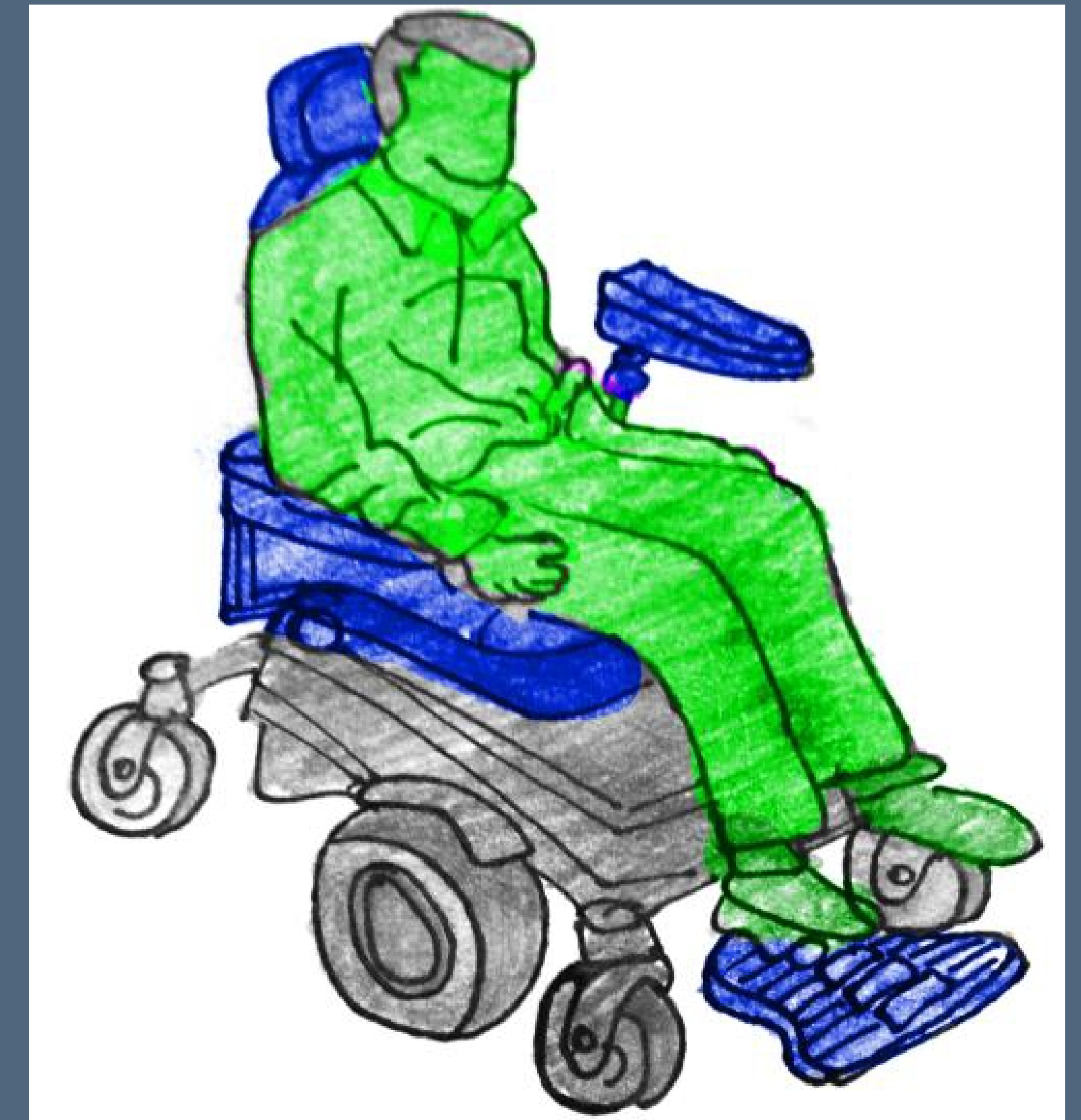
A “scrubbing”-style
interface for voiceover

Faster scrubbing invokes
more concise summaries

Expressivity and availability of gesture

[Carrington, Hurst, and Kane 2014]

From “wearables” to “chairables”:
integrating input and output
modalities on motor prostheses



Recognizing sign language

[Ma et al. 2018]

Could we unobtrusively
translate sign language?

Progress: by using the
ubicomp sensor pipeline via
wifi reflectance, researchers
can now detect over 250
different signs



Back to ability-based design

[Wobbrock et al. 2011; Wobbrock et al. 2018]

Core principles: focus on ability, not disability, respond to poor performance by changing systems, not users.

Recommended principles: self-adapt system or allow configuration, tailor support to user's performance and context, provide visibility and control over adaptations.

Encouraged: work with common commodity HW/SW

Benefits and gaps in this approach?

**Subtext beyond the
screen and individual**

DIY assistive tech

[Hurst and Tobias 2011; Hurst and Kane 2013; Buehler et al. 2015; Mankoff et al. 2019]

Many purchased assistive tools go unused, because they don't quite fit the needs of the individual

Could we empower more end users to tailor their own assistive devices, or those for friends and family?



Making for self and others

Many custom 3D printable designs were made on the behalf of oneself or for loved ones.
[Buehler et al. 2015]

This includes for those with disabilities, older adults, and those managing medications.

Often, designers lack formal training or expertise in assistive technology.

Example designs by inclusion criteria



thing:23729



thing:61329



thing:315819



thing:324072



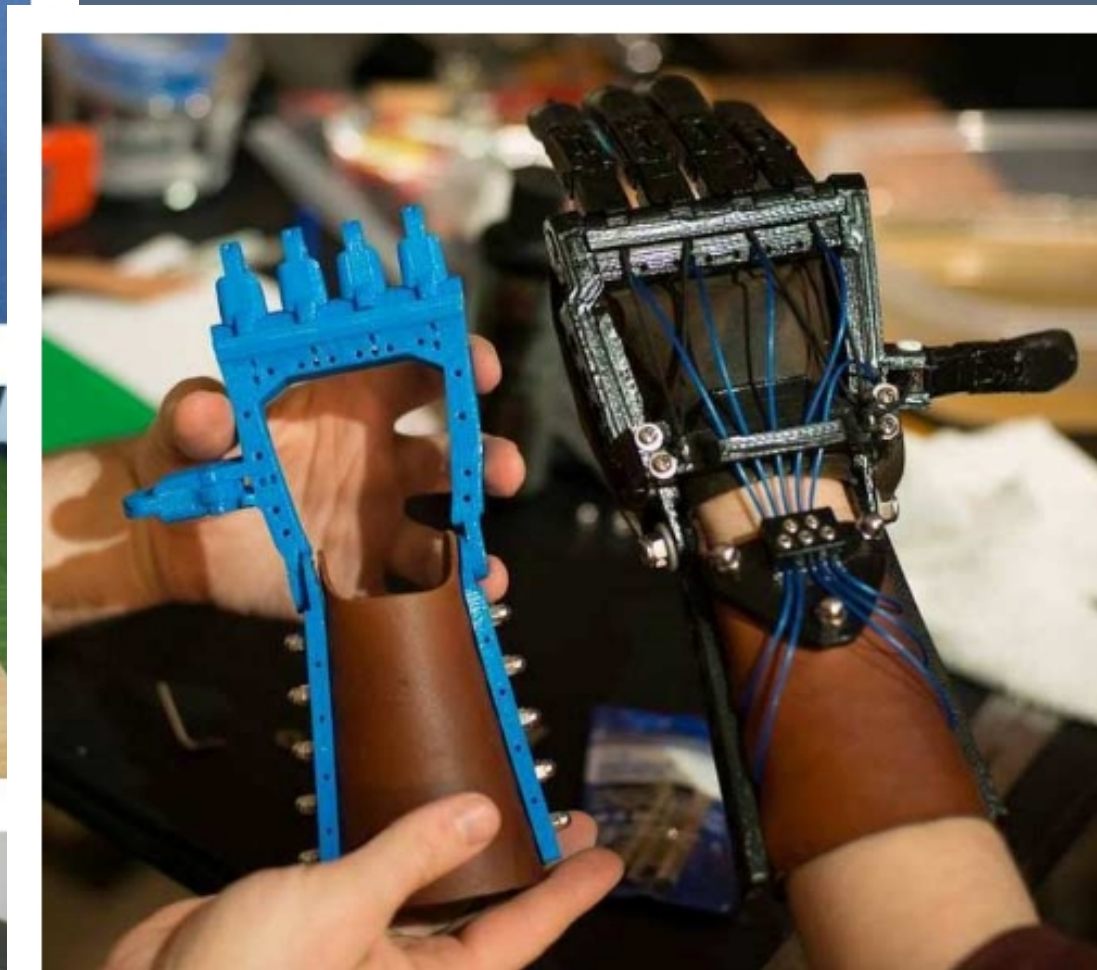
thing:261462



thing:84635



thing:359348



GesturePod

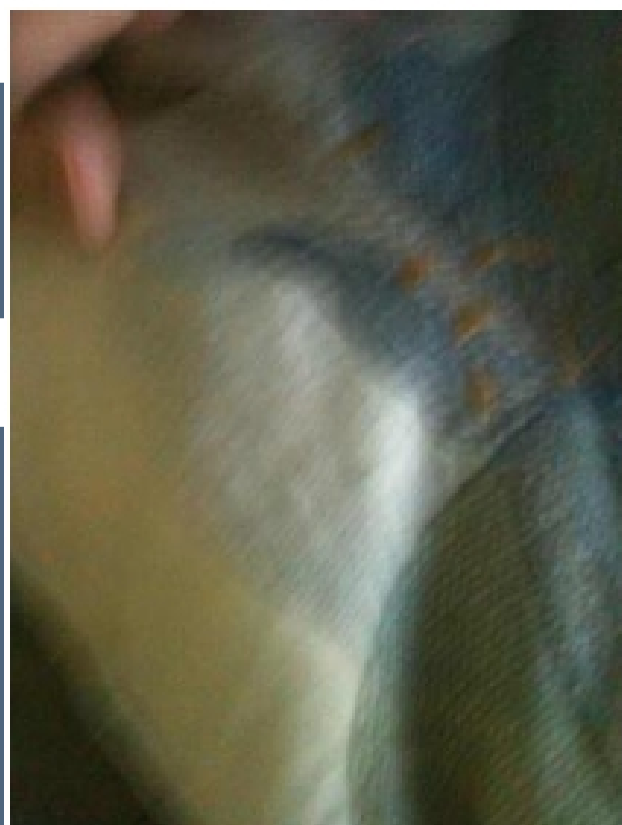
[Patil et al. UIST 2019]

Help on-demand anywhere

[Bigham et al., UIST '10]

VizWiz - Visual question answering for the blind via Amazon

What color is this pillow?



(89s) I can't tell.
(105s) multiple shades
of soft green, blue and
gold

What denomination is
this bill?



(24s) 20
(29s) 20

Do you see picnic tables
across the parking lot?



(13s) no
(46s) no

What temperature is my
oven set to?



(69s) it looks like 425
degrees but the image
is difficult to see.
(84s) 400
(122s) 450

Can you please tell me
what this can is?



(183s) chickpeas.
(514s) beans
(552s) Goya Beans

What kind of drink does
this can hold?

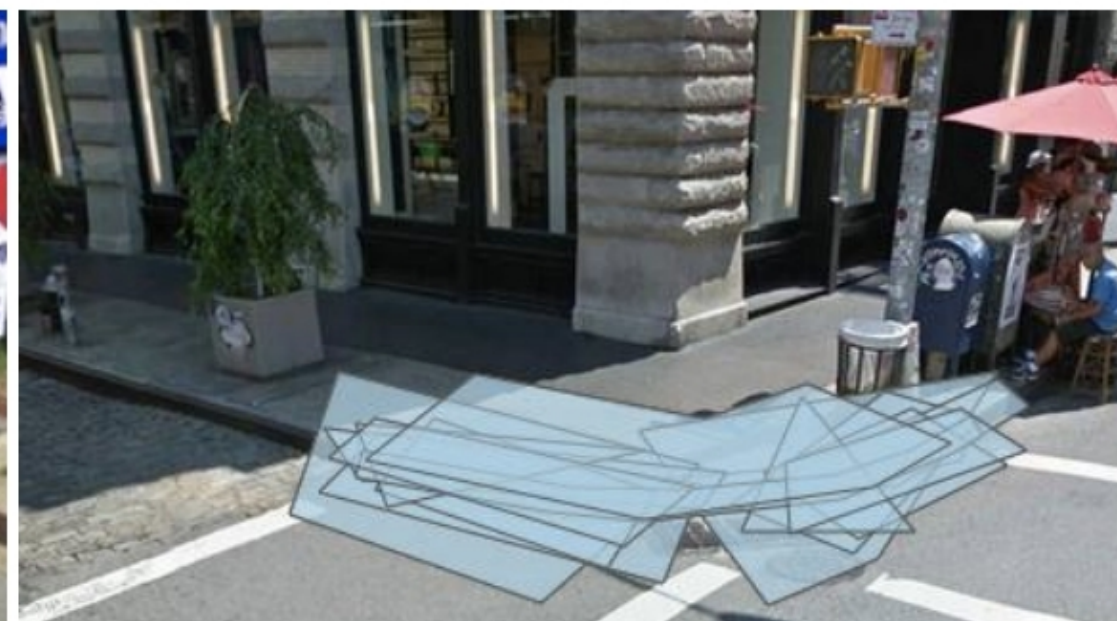


(91s) Energy
(99s) no can in the
picture
(247s) energy drink

Mapping out physical spaces

[Hara, Le, and Froehlich 2013]

Crowdsourced effort to label Google Street view images for accessibility issues



(a) *Object in Path*

(b) *No Curb Ramp*

(c) *Surface Problem*

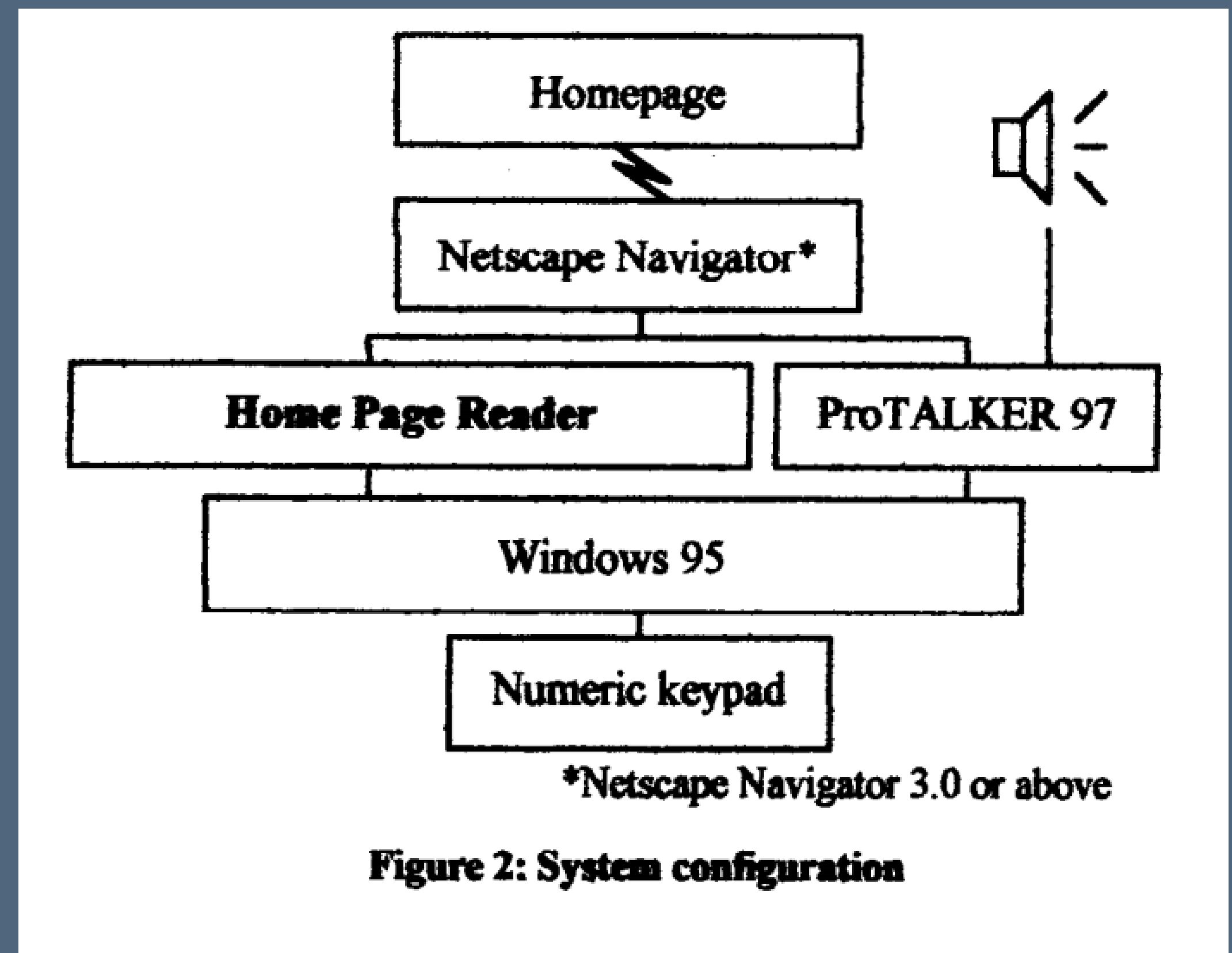
(d) *Prematurely Ending Sidewalk*

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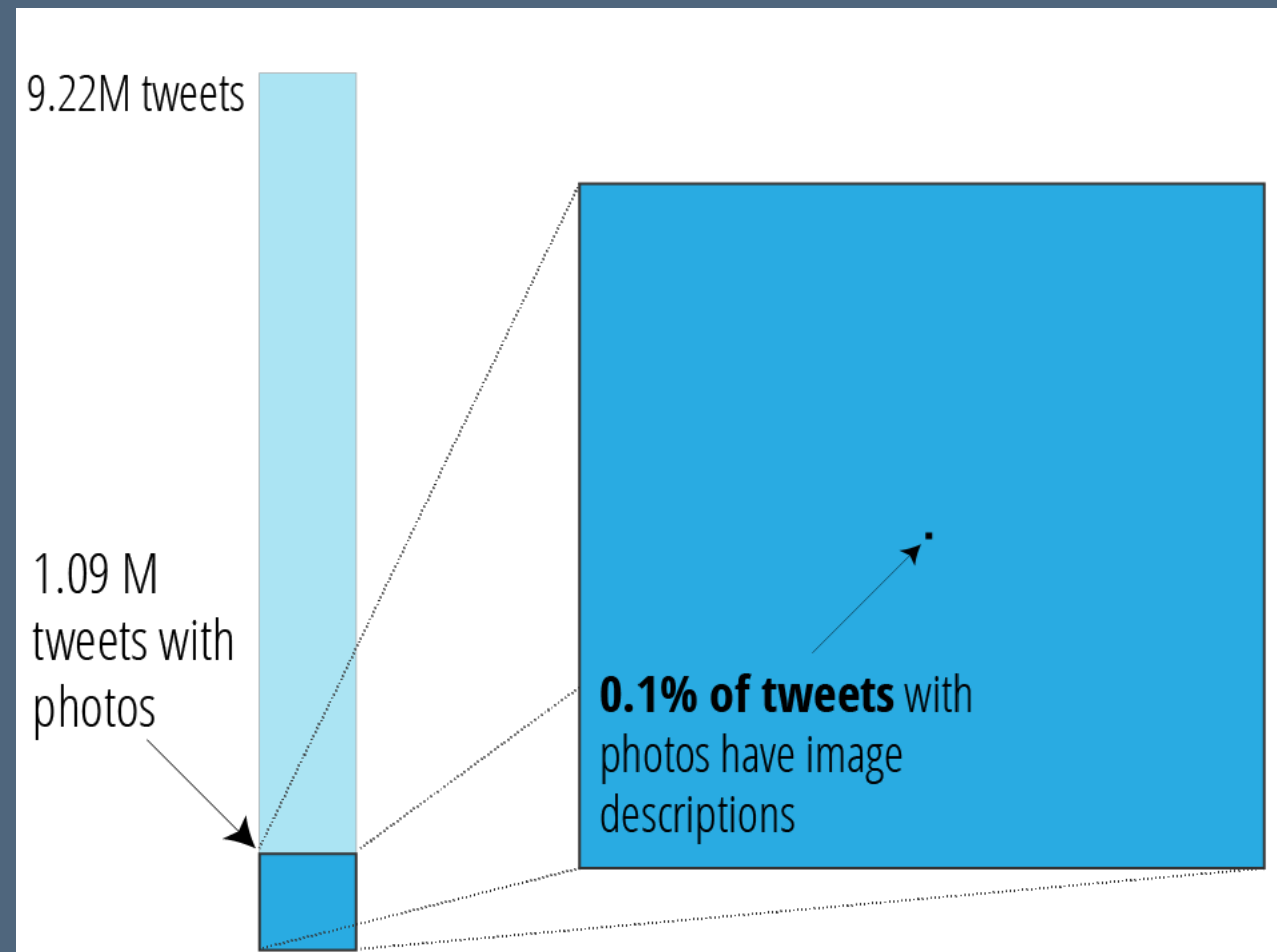


Social media: we have far to go

[Gleason et al. 2019]

Only 0.1% of images on social media contain user-contributed alt text

Even users who add alt text only add it half the time



**Benefiting accessibility
onto existing interfaces**

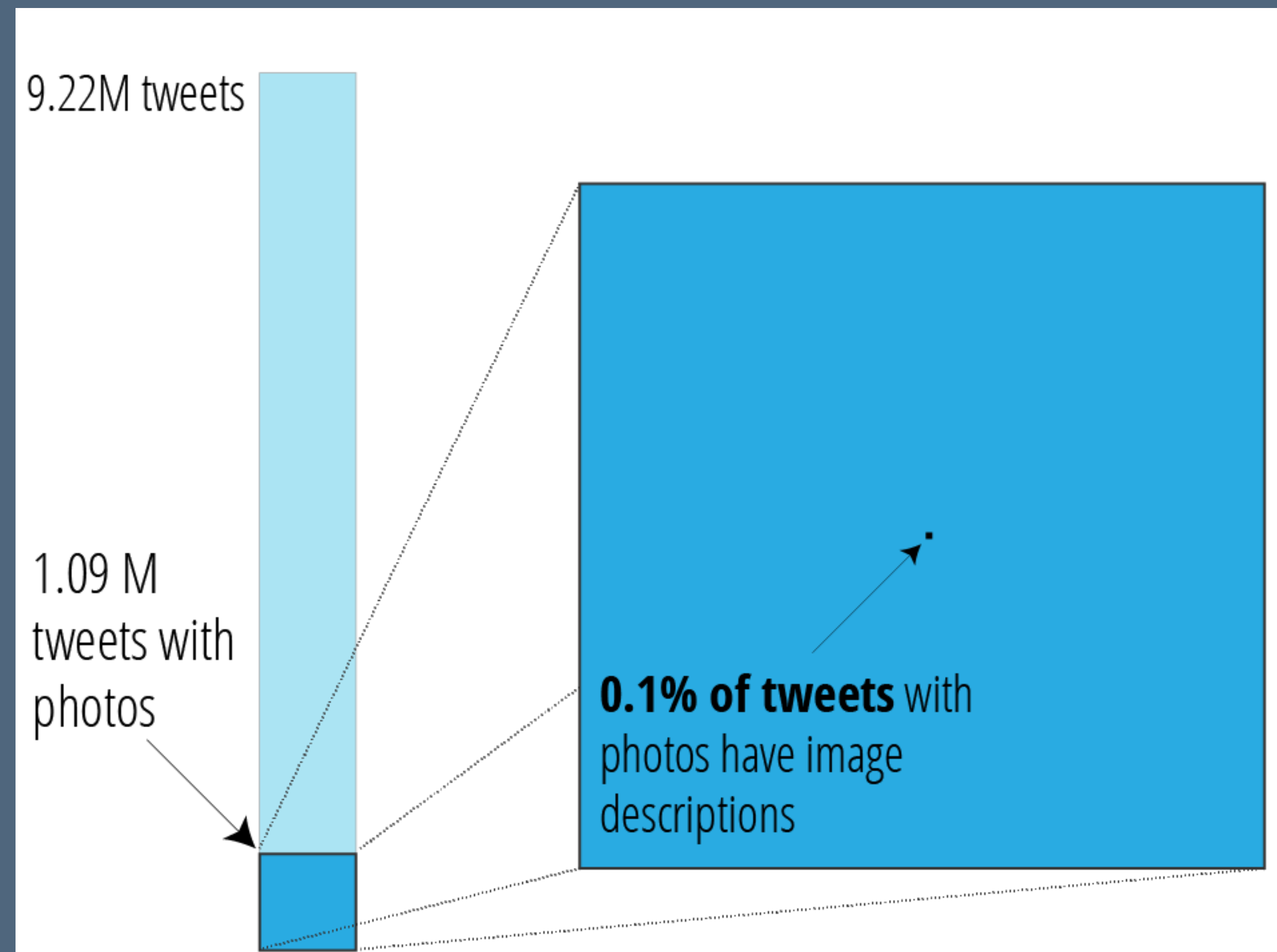
Failure to follow best practices at scale

[Gleason et al. 2019]

Only 0.1% of images on Twitter media contain user-contributed alt text

Even users who add alt text only add it half the time

On a sample of blind users' simulated feeds, only an estimated 5% of images had alt text.



Adding accessibility

AI-powered screen reader programming accessibility [Zhang et al.

iOS Screen Recognition



Making New AI fe

[Huh et al. 2023]

Prompt & Generated Images

A young chef cooks dinner for his parents.



Generate Questions
GPT-4

Prompt Verification Questions

Is the chef in the image?
How old is the young chef?
Are the parents visible in the image?

Answer
Questions
BLIP-2

Visual Content & Style Questions

What is the *setting* of this image?
What is the *emotion* in the image?
What is the *likely use* of this image?
What is the *medium* of this image?

(preset, informed by prompt guidelines)

BLIP-2
CLIP

Detect Objects
Detic

Image-Based Questions

What are they cooking?
What is the chef wearing?
Are cooking utensils visible?

BLIP-2

Caption BLIP-2
Generate Questions
GPT-4

Generate
Summary
GPT-4

GPT-4

Comparison Description

Similarities

All four images depict people cooking in a well-lit kitchen with happy expressions on their faces.

Differences

The differences among these images include medium (vector art, stock photo), focus on subject (a boy, a father, a mother) [...] Image 1 is a cartoon of a father and his children...

Per-Image Descriptions

In this stock photo, a young boy wears a chef's hat as he stands in a modern kitchen. He is preparing a salad using a knife...

2

Subtitle
Some additional thoughts

Many other areas, including:

Aging: e.g., can we support those with dementia to share their art with family? [Lazar, Edasis, and Piper 2017]

Neurodiversity: e.g., how might we craft interactive visual supports for communication between caregivers and children with autism? [Hayes et al. 2010]

Mental health

Chronic pain



Accessibility and disability

[Hofmann et al. 2020; Mankoff, Hayes, and Kasnitz 2010]

Accessibility: focus on creating technology related to impairment

Contributes an engineering lens

Disability studies: focus on understanding disability and advocating against ableism

Contributes a critical lens

These two lenses can be in tension: engineering often scopes down, but in the process, decontextualizes disability

How ought we to thread this needle?

3D-printed tactile picture books

[Stangl, Kim and Yeh 2014]

The pictures in
picture books
remain inaccessible
to blind children

3D printing might
allow us to make
books that convey
the contents of the
pictures in a tactile
channel

Summary

Ability-based design refocuses our attention from what a person cannot do to what they can do, and anchors the design process in their abilities

Accessibility sometimes requires going beyond supporting individual action to **community action**, and beyond screen interactions **adapting the physical world**.

Sensing and AI techniques can help us to **retrofit existing interfaces** to provide features that provide clear accessibility wins.

Questions about HCI &
Subtitle Email Lloyd May
Access! at lloyd.may@monash.edu

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